UniAgriQ

Agricultural Intelligence Platform

Genesis

Agricultural Investment Application

System Architecture Document

Technical Specification & Design Reference

Two Applications — One Unified Backend

AI-Driven · Satellite-Enhanced · IoT-Connected

Market-Intelligent · Tokenization-Ready

UniAgriQ — Farmer Intelligence **Genesis** — Agricultural Investment

Version 2.0

June 2026

Classification: Confidential — Patent & Engineering Reference

Prepared for internal review, patent filing, and engineering implementation.

Abstract

UniAgriQ is a mobile-first agricultural intelligence platform serving farmers through the **UniAgriQ** application and investors through the **Genesis** application. Both applications share a unified backend comprising AI engines, market intelligence, satellite analytics, IoT processing, and risk computation, while exposing fundamentally different user experiences tailored to their respective stakeholders.

UniAgriQ (Farmer App) integrates heterogeneous data sources — satellite remote sensing, IoT sensor networks, climate APIs, commodity market feeds, government agricultural databases, and structured farm reports — into an intelligence layer that drives crop recommendation, risk assessment, yield prediction, market timing, dynamic calendar generation, carbon credit estimation, and conversational advisory through a retrieval-augmented AI assistant named **Veda AI**.

Genesis (Investor App) reimagines agricultural investment as a premium fintech experience comparable to Zerodha, Groww, or Robinhood — except investments are made into diversified agricultural portfolios rather than listed equities. Genesis features **AI Portfolio Creation**, **Collective Investment**, **Land Tokenization**, a peer-to-peer marketplace, and transparent farm-level scoring backed by satellite validation and on-ground verification.

This document specifies the complete end-to-end system architecture including: a twelve-layer reference architecture, 18 detailed subsystem specifications, AI Portfolio Engine design, land tokenization workflow, collective investment module, multi-stage verification pipeline, marketplace engine, polyglot database architecture across seven storage engines, 30+ diagrams, formal scoring frameworks, security model with fraud detection, and a comprehensive observability stack.

Keywords: precision agriculture, agricultural intelligence, agricultural investment, land tokenization, fractional ownership, AI portfolio, crop recommendation, satellite remote sensing, IoT, market intelligence, collective investment, retrieval-augmented generation, carbon credit, risk engine, mobile application architecture.

Contents

1 Executive Summary

UniAgriQ addresses two fundamental market failures in agriculture:

1. **Farmer information asymmetry:** Farmers make high-stakes decisions — what to plant, when to irrigate, when to sell — with fragmented, stale, or absent data.
2. **Agricultural investment opacity:** Institutional and retail investors lack standardised, data-driven mechanisms to evaluate, diversify, and manage agricultural investments at farm-level granularity.

The platform resolves both through a **dual-application architecture** with a shared intelligent backend:

UniAgriQ (Farmer App) **ingests** multi-modal data from satellites, IoT sensors, climate APIs, and market feeds; **reasons** over it using ML models, rule engines, and scoring frameworks; and **delivers** actionable intelligence to farmers via mobile.

Genesis (Investor App) exposes the same intelligence layer to investors through a fintech-grade experience: AI-constructed diversified agricultural portfolios, collective investment vehicles, land tokenization with fractional ownership, a peer-to-peer marketplace, and transparent scoring backed by satellite validation.

The backend is designed as a **layered, event-driven, microservice-oriented** system with clear module boundaries. It is horizontally scalable, observable, and extensible to new crop types, geographies, and financial instruments without architectural rearchitecture.

2 Problem Statement

2.1 For Farmers

- **Information fragmentation:** Soil reports, weather forecasts, market prices, and government advisories exist in disconnected silos, often in incompatible formats.
- **Decision latency:** By the time a farmer learns of a pest outbreak, price crash, or weather anomaly, the optimal intervention window has passed.
- **Crop selection risk:** Farmers default to familiar crops, ignoring soil-specific suitability, climate trends, or emerging market demand.
- **Calendar rigidity:** Static region-wide calendars do not account for micro-climatic variation or multi-crop planning.
- **Market opacity:** Harvest-to-sell timing is distress-driven rather than demand-signal-driven.

2.2 For Investors

- **No standardised farm score:** No data-driven score exists to evaluate farm-level health, resilience, or growth potential.

- **No portfolio diversification tools:** Agricultural investment lacks the portfolio construction, diversification, and risk management tools available in equities and fixed income.
- **Illiquidity and opacity:** Agricultural assets are illiquid, non-fungible, and lack secondary markets for exit.
- **No fractional ownership:** Retail investors cannot participate in agriculture at ticket sizes below full land acquisition cost.
- **Verification gap:** No satellite-validated, AI-scored farm intelligence exists for investor consumption.

3 System Objectives

ID	Objective
O1	Provide a single-pane-of-glass farm health dashboard in UniAgriQ aggregating soil, climate, satellite, and sensor data.
O2	Generate personalised, multi-criteria crop recommendations ranked by composite suitability scores.
O3	Deliver real-time and forecast climate intelligence at farm-polygon resolution.
O4	Integrate commodity market, trade, and demand signals into a market intelligence engine with actionable sell-timing recommendations.
O5	Produce dynamic, farm-specific crop calendars.
O6	Compute satellite-derived vegetation indices with historical trend analysis and anomaly detection.
O7	Operate Veda AI as a retrieval-augmented conversational assistant.
O8	Quantify farm-level risk across eight dimensions.
O9	Estimate yield with confidence intervals and recommend optimal harvest windows.
O10	Estimate carbon credit potential per farm.
O11	Deliver a fintech-grade investor experience via Genesis with AI-constructed diversified agricultural portfolios.
O12	Enable collective investment into farms, FPOs, cooperatives, and agricultural projects.
O13	Implement land tokenization with fractional digital ownership, secondary market, and P2P transfers.
O14	Build a multi-stage verification pipeline for all stakeholders.
O15	Operate a production marketplace with listings, auctions, P2P trading, escrow, and settlement.
O16	Ensure zero hard-coded data: all outputs derived from live ingestion, model inference, or scheduled computation.
O17	Support horizontal scaling to 1 million farms and 500K investors without architectural change.

4 Platform Architecture Overview

4.1 Dual-Application, Shared-Backend Topology

UniAgriQ and Genesis are two independent mobile applications that share a common backend platform. Neither application communicates directly with the other; both interact exclusively through the unified API Gateway.

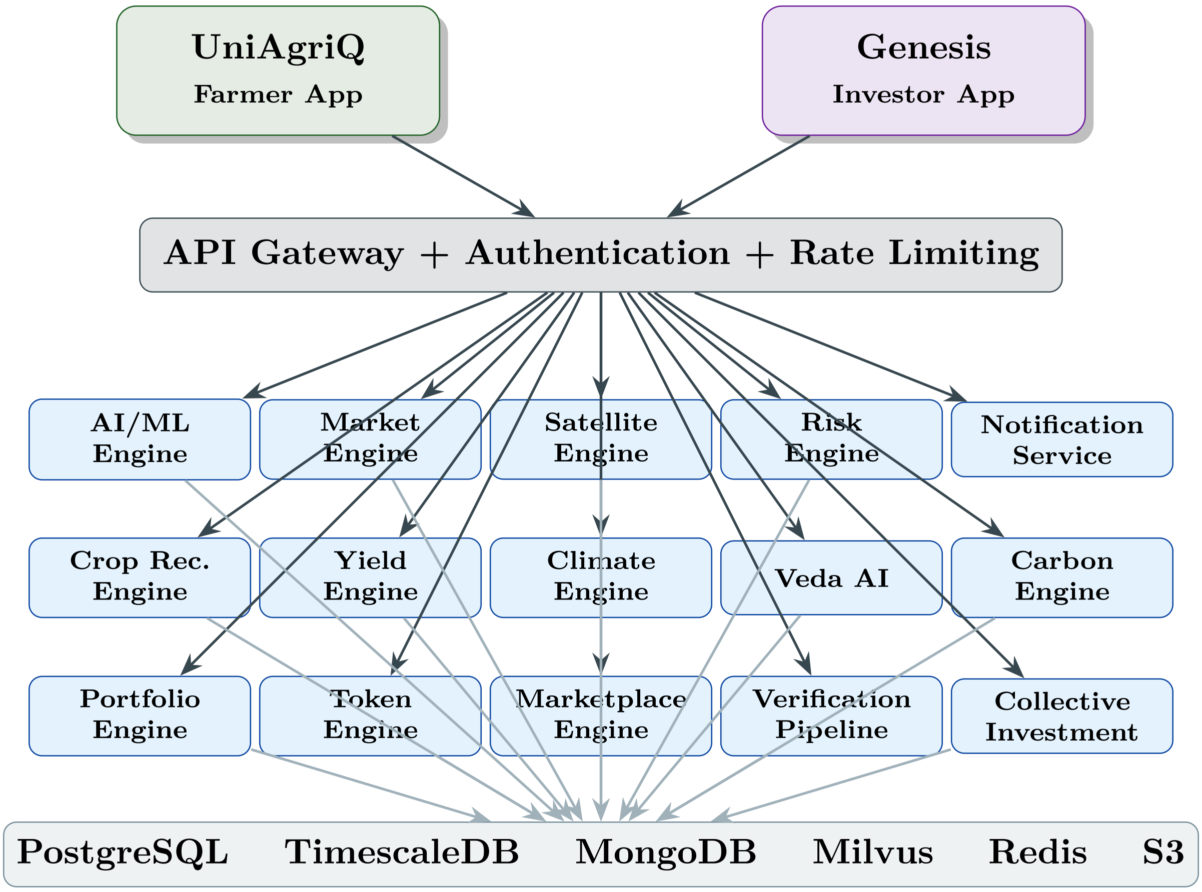

Figure 1: Dual-Application Architecture — UniAgriQ and Genesis share a unified backend.

4.2 Shared Backend Services

Both applications share these backend services:

- **API Gateway:** Authentication, authorisation, rate limiting, request routing. Role-based endpoint visibility: farmer endpoints invisible to investor tokens and vice versa.
- **AI/ML Engine:** Inference services, feature store, model registry. Serves both crop recommendation (UniAgriQ) and portfolio optimisation (Genesis).
- **Market Intelligence Engine:** Price trends, demand forecasts, opportunity alerts. Consumed by both farmer sell-timing and investor profit projections.
- **Satellite Engine:** NDVI, EVI, NDMI computation. Used for farm health (UniAgriQ) and farm validation/scoring (Genesis).
- **Risk Engine:** Multi-dimensional risk scoring. Powers farmer risk dashboard and investor portfolio risk assessment.

- **Authentication:** OAuth 2.0 + JWT. Shared identity provider with role-based access.
- **Notification Service:** Push, SMS, in-app, email. Shared infrastructure with role-specific routing.
- **Database Layer:** Polyglot persistence shared across both applications with logical schema separation.

5 Functional Requirements

5.1 UniAgriQ — Farmer Application

5.1.1 Farm Profile & Onboarding (FR-100)

- FR-101: Register farmer identity with KYC-light verification.
- FR-102: Register farms with geofenced polygon boundaries.
- FR-103: Record crop history per farm.
- FR-104: Upload soil/farm reports with automated extraction.
- FR-105: Capture ownership/tenancy/cooperative metadata.
- FR-106: Multi-farm management under single identity.

5.1.2 Farm Health Dashboard (FR-200)

- FR-201: Composite Farm Health Score with colour-coded status (red/blue/green).
- FR-202: Current crop, growth stage, multi-crop summary.
- FR-203: Farm polygon on satellite map with NDVI overlay.
- FR-204: Real-time sensor summaries.
- FR-205: Upcoming activity timeline from crop calendar.
- FR-206: Advisory feed ranked by urgency.
- FR-207: Alert banner for critical events.
- FR-208: Opportunity alert cards.

5.1.3 Farm Report Intake (FR-300)

- FR-301: Accept soil report via PDF, camera, or manual entry.
- FR-302: Extract and normalise: soil type, pH, N-P-K, organic carbon, macro/micro nutrients, EC, moisture.
- FR-303: Validate against agronomic reference ranges.

- FR-304: Generate visual soil health dashboard with radar charts.
- FR-305: Store structured report with farm linkage.

5.1.4 Climate Intelligence (FR-400)

- FR-401: Current conditions at farm centroid.
- FR-402: 7-day and 30-day forecast with confidence bands.
- FR-403: Heat stress and drought stress indicators.
- FR-404: 10-year historical climate data.
- FR-405: Climate fitness score per crop.

5.1.5 Crop Recommendation (FR-500)

- FR-501: Ranked list of up to 15 candidate crops.
- FR-502: Multi-criteria scoring (soil, climate, rainfall, market, risk).
- FR-503: Reject below configurable threshold (default 0.45).
- FR-504: Classify into primary, secondary, fast-turnaround, resilience, balancing, and fencing/protective.
- FR-505: Reasoning summary per recommendation.

5.1.6 Market Intelligence (FR-600)

- FR-601: Mandi/APMC prices, trading signals, import-export trends.
- FR-602: Profit score and market timing score per crop.
- FR-603: Harvest-to-sell recommendation with optimal window.
- FR-604: Market opportunity alerts.
- FR-605: Month-wise selling calendar.

5.1.7 Dynamic Crop Calendar (FR-700)

- FR-701: Month-wise activity calendar for active crops.
- FR-702: Sowing, irrigation, fertiliser, pest control, harvest.
- FR-703: Priority colour-coding (critical/important/routine).
- FR-704: Dynamic adjustment based on weather updates.
- FR-705: Multi-crop overlapping calendar support.

5.1.8 Satellite Intelligence (FR-800)

- FR-801: NDVI, EVI, NDMI at 5-day cadence.
- FR-802: 10-year historical index time series.
- FR-803: Anomaly detection against historical baseline.
- FR-804: Drought and flood scores.
- FR-805: Seasonal trend reports.

5.1.9 Veda AI Chatbot (FR-900)

- FR-901: Natural language queries in English and regional languages.
- FR-902: Farm-context-grounded responses.
- FR-903: RAG retrieval from knowledge base.
- FR-904: Action-oriented responses with deep-links.
- FR-905: Proactive urgent notifications.

5.1.10 Risk, Yield, Carbon (FR-1000–1200)

- FR-1001: 8-dimension risk scoring with alerts.
- FR-1101: Yield prediction with confidence interval.
- FR-1102: Smart harvest window recommendation.
- FR-1201: Carbon sequestration estimate mapped to credit frameworks.

5.2 Genesis — Investor Application

5.2.1 Investor Onboarding & Verification (FR-2000)

- FR-2001: Register investor with full KYC (PAN, Aadhaar, bank).
- FR-2002: AML screening and risk profile assessment.
- FR-2003: Income verification and investment eligibility check.
- FR-2004: Investor wallet creation with escrow integration.

5.2.2 Genesis Home Dashboard (FR-2100)

- FR-2101: Portfolio value, today's gain/loss, expected annual return.
- FR-2102: Risk score, diversification score, farm health index.
- FR-2103: Carbon credit value, ESG/sustainability score.
- FR-2104: Weather exposure, global market exposure.
- FR-2105: AI suggestions, new investment opportunities.
- FR-2106: Portfolio performance graph, allocation pie chart.
- FR-2107: Recent transactions, pending verifications.
- FR-2108: Harvest timeline, profit calendar, exit opportunities.
- FR-2109: P2P marketplace activity, trending crops.
- FR-2110: Investment wallet balance and transaction history.

5.2.3 AI Portfolio Engine (FR-2200)

- FR-2201: Accept investor inputs: amount, risk appetite, preferred states, preferred crops, horizon, liquidity, ESG preference.
- FR-2202: AI constructs diversified agricultural portfolio balancing crop, geographic, climate, harvest season, market, risk, water, and disease diversity.
- FR-2203: Display portfolio composition with farm cards, allocation weights, expected return, risk metrics.
- FR-2204: Rebalancing recommendations on material change.
- FR-2205: Multiple portfolio templates (conservative, balanced, growth, ESG-focused, high-yield).

5.2.4 Collective Investment (FR-2300)

- FR-2301: Create collective investment vehicles for farms, FPOs, cooperatives, warehouses, processing units.
- FR-2302: Multiple investors contribute to single vehicle.
- FR-2303: AI allocates ownership, profit share, risk contribution, voting rights, CAP table.
- FR-2304: Track expected ROI, actual distributions, exit options.

5.2.5 Land Tokenization (FR-2400)

- FR-2401: Digitise approved agricultural assets into ownership tokens.
- FR-2402: Multi-stage verification: land records, survey number, geofence matching, ownership, KYC, AML, legal due diligence.
- FR-2403: Fractional unit creation and digital ownership certificates.
- FR-2404: Secondary market listing and P2P transfers.
- FR-2405: Escrow-based settlement and transaction ledger.
- FR-2406: Exit mechanism with token burn.

5.2.6 Marketplace (FR-2500)

- FR-2501: Farm and investment listings with search and filters.
- FR-2502: Auction mode, fixed price, and instant buy.
- FR-2503: Watchlist and wishlist.
- FR-2504: Order matching and settlement with escrow.
- FR-2505: Fees, taxes, ownership transfer, portfolio update.

6 Non-Functional Requirements

Category	Requirement	Target
Performance	Dashboard API p95 latency	$< 500\,\mathrm{ms}$
Performance	Veda AI end-to-end response	$< 3\,\mathrm{s}$
Performance	Portfolio construction	$< 15\,\mathrm{s}$
Scalability	Registered farms	1M
Scalability	Registered investors	500K
Scalability	Concurrent mobile users	100K peak
Availability	Platform uptime	99.9%
Security	Data at rest	AES-256
Security	Data in transit	TLS 1.3
Security	Authentication	OAuth 2.0 + JWT
Security	Financial transactions	PCI-DSS aligned
Compliance	Data residency	India-region cloud

7 High-Level System Architecture

UniAgriQ follows a **twelve-layer reference architecture**.

Figure 2: UniAgriQ Twelve-Layer Reference Architecture.

Layer 5 — Investment Layer is new and contains the Portfolio Engine, Tokenization Engine, Collective Investment Module, and Marketplace Engine. **Layer 11 — Financial Layer** handles escrow, settlement, wallet management, and the immutable transaction ledger required for tokenized asset trading.

8 Detailed Subsystem Architecture

Each subsystem is specified with: *Purpose*, *Inputs*, *Outputs*, *Internal Logic*, *Dependencies*, *Storage*, *API Dependencies*, *Model Dependencies*, *Failure Modes*, and *Update Cadence*.

8.1 Farm Profile & Onboarding Service

Attribute	Description

Purpose	Register and maintain farmer identities, farm polygons, crop histories, ownership, and report linkages.
Inputs	Registration form, GPS/polygon vertices, crop history, soil report uploads, KYC documents.
Outputs	Stateful farm profile (farmer_id, farm_id, polygon GeoJSON, crop history, linked reports, ownership metadata).
Internal Logic	1) Validate phone via OTP. 2) Geocode address. 3) Accept polygon drawing or auto-detect from cadastral overlay. 4) Hash Aadhaar. 5) Link reports. 6) Compute area via PostGIS `ST_Area`.
Storage	PostgreSQL (`farmers`, `farms`, `crop_history`), PostGIS (`farm_polygons`).
Failure Modes	OTP timeout (retry with backoff), polygon self-intersection (client-side validation).
Update Cadence	Event-driven (on user action).

8.2 Farm Report Intake Module

Attribute	**Description**
Purpose	Extract, normalise, validate, and store structured soil data from heterogeneous inputs.
Inputs	PDF report, camera image, or manual form.
Outputs	Structured record: soil_type, pH, N, P, K, organic_carbon, Fe, Zn, Mn, Cu, B, S, Ca, Mg, EC, moisture, confidence.
Internal Logic	1) OCR via Google Vision API → NER extraction. 2) Schema validation. 3) Unit normalisation. 4) Flag 3σ outliers. 5) Generate radar chart payload.
Storage	MongoDB (`soil_reports`), S3 (raw artefacts).
Model Dependencies	Soil Report NER (custom PyTorch).
Failure Modes	OCR confidence < 0.6 (fallback to manual entry).
Update Cadence	Event-driven (on upload).

8.3 Climate Intelligence Engine

Attribute	**Description**
Purpose	Real-time, forecast, and historical climate data at farm-polygon resolution; derived stress indicators.
Inputs	Farm centroid (lat, lon), date range, crop growth stage.
Outputs	Current conditions, 7/30-day forecast, historical normals, HSI, DSI, climate fitness score per crop.
Internal Logic	1) Fetch from weather API. 2) Historical normals from ERA5/IMD. 3) HSI $= f(\text{max\_temp}, \text{duration}, \text{crop\_stage})$. 4) DSI from SPEI. 5) Climate fitness: Gaussian fit against crop-optimal ranges.
Storage	TimescaleDB (`climate_observations`, `climate_forecasts`).
API Dependencies	OpenWeatherMap, IMD API, Tomorrow.io.

Failure Modes	Primary API timeout (failover to secondary).
Update Cadence	Current: 15 min. Forecast: 6 hr. Historical: monthly.

8.4 Crop Recommendation Engine

Attribute	Description
Purpose	Ranked, multi-criteria crop recommendations personalised to farm soil, climate, and market context.
Inputs	Soil report, farm polygon, climate profile, rainfall history, market signals, farmer preferences.
Outputs	Up to 15 crops with composite score, sub-scores, class label (primary/secondary/fast-turnaround/resilience/balancing/fencing), reasoning summary.
Internal Logic	1) Candidate retrieval from crop knowledge base (region + season). 2) Compute: S_s (soil), S_c (climate), S_r (rainfall), S_m (market), S_d (disease risk — inverted). 3) Composite: $S = 0.25S_s + 0.20S_c + 0.15S_r + 0.25S_m - 0.15S_d$. 4) Reject $S < 0.45$. 5) Assign class labels. 6) Generate reasoning.
Model Dependencies	Crop suitability (XGBoost), market forecaster.
Failure Modes	Missing soil report (use regional defaults with reduced confidence).
Update Cadence	On-demand + nightly batch.

8.5 Market Intelligence Engine

Attribute	Description
Purpose	Aggregate and forecast agricultural commodity market signals; generate profit scores, sell-timing, and opportunity alerts.
Inputs	Mandi/APMC feeds, commodity exchange, import-export stats, Google Trends, news sentiment, festival calendar.
Outputs	Profit score, market timing score, selling calendar, opportunity alerts, demand forecast curves.
Internal Logic	1) Price trend analysis (30/60/90-day MA, volatility). 2) Demand signal aggregation (festival 0.2, search 0.15, news 0.15, seasonal 0.5). 3) Supply gap detection (production vs. demand; flag >10%). 4) Profit score: $P = \text{norm(price)} \times \text{demand} \times (1 - \text{volatility})$. 5) Sell calendar: months with price $\geq$ 85th percentile. 6) Opportunity alerts: rule-based triggers.
Model Dependencies	Demand Forecaster (LSTM/Prophet), News Sentiment (fine-tuned BERT).
Update Cadence	Prices: hourly. Forecast: daily. Alerts: event-driven.

8.6 Dynamic Crop Calendar Generator

Attribute	Description
Purpose	Farm-specific, weather-adaptive crop activity calendar.
Inputs	Selected crops, sowing date, farm location, soil report, climate forecast, crop knowledge base, NDVI trend.
Outputs	Month-wise activities with task type, date window, priority (critical/important/routine), input quantity.
Internal Logic	1) Load crop template. 2) Adjust sowing for rainfall forecast. 3) Shift irrigation based on precipitation. 4) Advance fertiliser based on NDVI growth stage. 5) Trigger pest control when disease risk > 0.6. 6) Harvest window from GDD accumulation. 7) Multi-crop conflict resolution.
Failure Modes	Weather unavailable (use climatological normals).
Update Cadence	On crop selection, weather change, weekly batch.

8.7 Satellite Intelligence Engine

Attribute	Description
Purpose	Vegetation health, moisture, and anomaly metrics from remote sensing at farm-polygon resolution.
Inputs	Farm polygon (GeoJSON), Sentinel-2/Landsat imagery.
Outputs	NDVI, EVI, NDMI, drought score, flood score, crop consistency index, vegetation trend, anomaly flags, 10-year history.
Internal Logic	1) Query satellite provider for cloud-free scenes. 2) Atmospheric correction. 3) Pixel-level indices: NDVI = $(NIR - RED)/(NIR + RED)$, EVI = $2.5(NIR - RED)/(NIR + 6RED - 7.5BLUE + 1)$, NDMI = $(NIR - SWIR)/(NIR + SWIR)$. 4) Polygon aggregation. 5) Historical comparison ($\|z\| > 2.0$ = anomaly). 6) Drought/flood scoring. 7) Spatial CV for consistency.
API Dependencies	Sentinel Hub / Google Earth Engine.
Failure Modes	Persistent cloud cover (interpolate from adjacent dates).
Update Cadence	Every 5 days (Sentinel-2 revisit).

8.8 Veda AI — Retrieval-Augmented Farm Assistant

Attribute	Description
Purpose	Context-aware agricultural conversational assistant grounded in structured knowledge and farm data.
Inputs	User query (text/voice), conversation history, farm profile context.
Outputs	Natural language response with optional action card, deep-link, confidence indicator.

Internal Logic	1) Intent classification. 2) Context injection (farm, crop, soil, season). 3) Vector retrieval (top-5 chunks, similarity > 0.72). 4) Structured data retrieval from engine APIs. 5) LLM response generation with factuality enforcement. 6) Hallucination guard. 7) Proactive alert monitoring.
Storage	Milvus (`knowledge_embeddings`), PostgreSQL (`chat_sessions`), Redis (session cache).
Model Dependencies	Embedding model, LLM, intent classifier, hallucination detector.
Failure Modes	LLM timeout (cached Q&A fallback), vector store down (keyword search fallback).
Update Cadence	Knowledge re-indexed weekly. Live data per query.

8.9 Risk Engine

Attribute	**Description**
Purpose	Multi-dimensional risk scores with trend analysis and alerts.
Inputs	Climate, satellite, soil, market, IoT sensors, pest data, trade data.
Outputs	Composite risk (0–1), per-dimension scores (weather, disease, nutrient, water, economic, market volatility, pest, logistics), trend arrows, triggered alerts.
Internal Logic	Eight independent dimension scorers → weighted aggregation (configurable, default 1/8 each) → trend comparison (7/30-day MA) → alert triggers on threshold breach.
Failure Modes	Missing dimension data (partial score with reduced confidence).
Update Cadence	Hourly (weather), daily (market, nutrient).

8.10 Yield & Harvest Prediction Engine

Attribute	**Description**
Purpose	Predict crop yield with confidence intervals; recommend optimal harvest timing.
Inputs	Soil, climate, NDVI time series, crop variety, irrigation, input log, historical yields.
Outputs	Predicted yield (tonnes/ha) with 90% CI, yield confidence score, harvest date range, performance trend, loss risk %.
Internal Logic	Feature construction → yield regression (XGBoost) with quantile regression (10th/50th/90th) → harvest timing from NDVI plateau + GDD maturity → Monte Carlo loss simulation.
Model Dependencies	Yield regression, quantile regression, NDVI trajectory model.
Update Cadence	Weekly during growing season.

8.11 Carbon Credit Estimation Engine

Attribute	Description
Purpose	Farm-level carbon sequestration estimate mapped to credit frameworks.
Inputs	Farm area, SOC, crop type, residue management, tillage, irrigation, fertiliser, cover crop, agroforestry.
Outputs	Annual sequestration (tCO_2e/ha), carbon score, eligible framework (Verra/Gold Standard), credit value, practice recommendations.
Internal Logic	IPCC Tier 2 stock-change factors → management practice multipliers → aggregate C pools → normalise against regional benchmarks → map to credit value.
Failure Modes	Missing practice data (assume conventional with disclaimer).
Update Cadence	On practice change; annual batch.

8.12 IoT Sensor Intelligence Module

Attribute	Description
Purpose	Ingest, validate, and process real-time farm sensor streams.
Inputs	Soil moisture, temperature, humidity, leaf wetness, rain gauge, light intensity, wind speed.
Outputs	Validated time series, anomaly flags, farm-level summary, irrigation trigger recommendation.
Internal Logic	MQTT ingestion → schema validation → z-score anomaly detection → gap interpolation → irrigation trigger (moisture $<$ MAD for $>$2 readings) → spatial aggregation.
Storage	TimescaleDB (`sensor_readings`), Redis (latest cache).
Failure Modes	Sensor offline >4h (alert farmer, exclude).
Update Cadence	Real-time (sub-minute).

9 Genesis — Investment Subsystem Architecture

9.1 AI Portfolio Engine

Attribute	Description
Purpose	Construct diversified agricultural portfolios analogous to mutual funds or ETFs, where the underlying assets are farm-level investment positions instead of listed equities.
Inputs	Investment amount, risk appetite (1–10), preferred states, preferred crops, investment horizon (months), liquidity preference (lock-in tolerance), ESG preference (0–1).
Outputs	Portfolio specification: list of farm positions with per-farm allocation weight, expected return, risk contribution, harvest timeline, satellite health score, diversification metrics.

Internal Logic	**Step 1 — Universe Construction.** Filter eligible farms by: verification status = approved, investor score > 0.50, state preference match, crop preference match. **Step 2 — Feature Extraction.** For each candidate farm, extract: yield confidence, risk score, climate resilience, satellite health trend, market demand score, carbon score, water dependency, disease exposure, insurance coverage, historical yield variance, harvest month. **Step 3 — Diversity Scoring.** Compute pairwise correlation matrix across farms on dimensions: crop type, geographic region, climate zone, harvest month, water source, disease profile. Penalise correlated pairs. **Step 4 — Portfolio Optimisation.** Solve constrained optimisation: $$\max_{\mathbf{w}} \ \mathbf{w}^T\boldsymbol{\mu} - \lambda\mathbf{w}^T\boldsymbol{\Sigma}\mathbf{w} + \gamma D(\mathbf{w})$$ subject to $\sum w_i = 1$, $w_i \geq 0$, $w_i \leq w_{\max}$ (concentration limit), and liquidity/horizon constraints. Here $\boldsymbol{\mu}$ is expected return vector, $\boldsymbol{\Sigma}$ is risk covariance, $D(\mathbf{w})$ is a diversity bonus, λ maps to risk appetite, γ controls diversity weight. **Step 5 — Template Matching.** If investor selects a template (conservative, balanced, growth, ESG, high-yield), override λ and γ with preset values and apply additional constraints (e.g., ESG template: carbon score > 0.7, no chemical-intensive crops). **Step 6 — Rebalancing.** Monitor portfolio monthly. Trigger rebalancing when: (a) any farm's risk score increases >20%, (b) satellite anomaly detected, (c) harvest completed, (d) new higher-opportunity farm enters universe.
Dependencies	Risk Engine, Satellite Engine, Market Engine, Yield Engine, Carbon Engine, Verification Pipeline.
Storage	PostgreSQL (`portfolios`, `portfolio_positions`, `portfolio_history`).
Model Dependencies	Portfolio optimiser (scipy / cvxpy), return predictor, risk covariance estimator.
Failure Modes	Insufficient eligible farms (widen filters with user notification), optimisation infeasible (relax constraints incrementally).
Update Cadence	On creation; monthly rebalancing review; event-driven on material risk change.

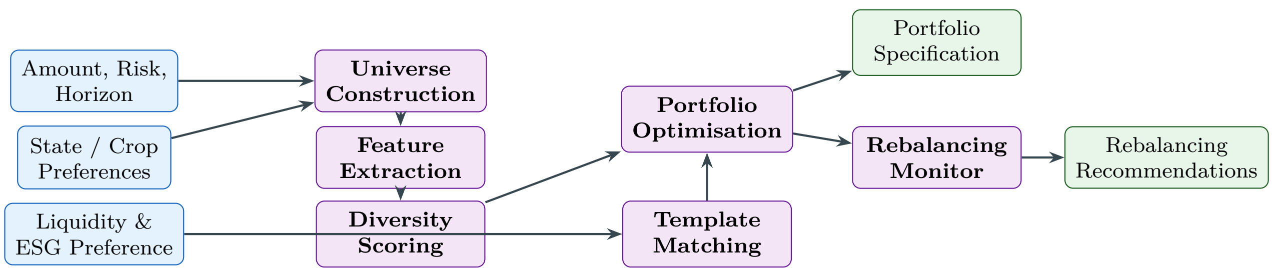

Figure 3: AI Portfolio Engine — Construction and Rebalancing Pipeline.

9.2 Collective Investment Module

Attribute	Description
Purpose	Enable multiple investors to jointly invest in a single agricultural asset (farm, FPO, cooperative, warehouse, processing unit) with automated ownership, profit sharing, and governance allocation.
Inputs	Asset listing (verified farm/FPO/project), investor contributions, investment terms (lock-in, exit rules, profit distribution frequency).
Outputs	Collective vehicle record: CAP table (investor → ownership %), profit distribution schedule, voting weight matrix, expected ROI, actual distribution history, exit calculations.
Internal Logic	**Vehicle Creation:** Asset owner or platform creates a collective vehicle with: target raise, minimum ticket, maximum investors, lock-in period, profit distribution model (pro-rata or performance-tiered), governance model (voting rights proportional to ownership or one-investor-one-vote). **Subscription:** Investors subscribe with amount → escrow holds funds → on target raise completion, escrow releases to asset owner → CAP table minted. **Ownership Allocation:** $\text{ownership}_i = \text{contribution}_i/\text{total\_raise}$. **Profit Distribution:** On harvest revenue or periodic audit: compute net profit = revenue − costs − platform_fee; distribute $\text{profit}_i = \text{ownership}_i \times \text{net\_profit}$. **Risk Contribution:** $\text{risk}_i = \text{ownership}_i \times \text{asset\_risk\_score}$. **Voting:** Governance decisions (extend lock-in, approve additional investment, exit decision) require weighted majority. **Exit:** Investor can exit via: (a) secondary market sale of ownership stake, (b) scheduled exit window after lock-in, (c) asset liquidation. Exit value computed from latest asset valuation minus exit penalty if before maturity.
Dependencies	Verification Pipeline, Escrow Service, Marketplace, Tokenization Engine.
Storage	PostgreSQL (`collective_vehicles`, `cap_table`, `distributions`, `governance_votes`).
Failure Modes	Target raise not met within window (return funds from escrow), asset owner default (trigger dispute resolution).
Update Cadence	Event-driven (on subscription, distribution, vote).

9.3 Land Tokenization Engine

Attribute	Description
Purpose	Digitise approved agricultural assets into fractional ownership units with full lifecycle management: creation, trading, transfer, and retirement.

Tokenization Lifecycle:

Phase 1 — Asset Verification. Multi-stage verification before any asset is eligible for tokenization: (a) Government land record validation (7/12 extract, patta, khatauni). (b) Survey number cross-check against cadastral database. (c) GeoFence polygon matching: uploaded polygon compared against government survey boundaries using PostGIS `ST_Intersection`; match threshold >85%. (d) Ownership verification: title search, encumbrance check, lien check. (e) KYC of asset owner. (f) AML screening. (g) Legal due diligence: third-party legal opinion on clear title. (h) Satellite validation: confirm agricultural use via NDVI > 0.3 over trailing 12 months. (i) Optional physical inspection by platform-empanelled inspector.

Phase 2 — Token Creation. On successful verification: (a) Asset registered in `token_registry` with unique asset_id. (b) Total ownership denominated into N fractional units (configurable; default $N = 10{,}000$ per hectare). (c) Each unit receives a unique token_id linked to asset_id. (d) Digital Ownership Certificate generated (PDF with QR code linking to on-chain/on-ledger record). (e) Smart Ownership Ledger entry: immutable append-only log recording creation event with timestamp, verifier_id, and evidence hashes.

Phase 3 — Marketplace Listing. Tokens listed on platform marketplace with: price per unit, total units available, asset metadata, satellite thumbnail, verification badge, yield forecast, risk score.

Phase 4 — Trading & Transfer. (a) **Primary sale:** Asset owner sells initial units to investors. (b) **Secondary market:** Investors list owned units for resale. (c) **P2P transfer:** Direct transfer between verified investors. All transactions use escrow: buyer funds held in escrow → ownership transferred in token_registry → funds released to seller → transaction recorded in immutable ledger.

Phase 5 — Settlement & Audit. Settlement engine: T+1 settlement for token trades. Ownership history maintained as append-only chain: each transfer creates a new entry with prev_hash linking to prior owner. Audit trail: every state change (creation, transfer, dividend, burn) logged with actor, timestamp, evidence hash.

Phase 6 — Exit & Token Burn. On asset exit (sale of underlying land, project completion): (a) Compute exit value per unit from asset valuation. (b) Distribute exit proceeds pro-rata. (c) Burn all tokens: mark status = BURNED in registry, preventing further trading. (d) Final audit entry recording burn event.

Fraud Detection: (a) Duplicate asset detection: polygon overlap check against all registered assets (overlap $> 20\% \Rightarrow$ flag). (b) Ownership chain validation: verify unbroken chain from creation. (c) Velocity checks: flag if same token transferred >3 times in 24 hours. (d) Price anomaly: flag if trade price deviates >50% from estimated fair value.

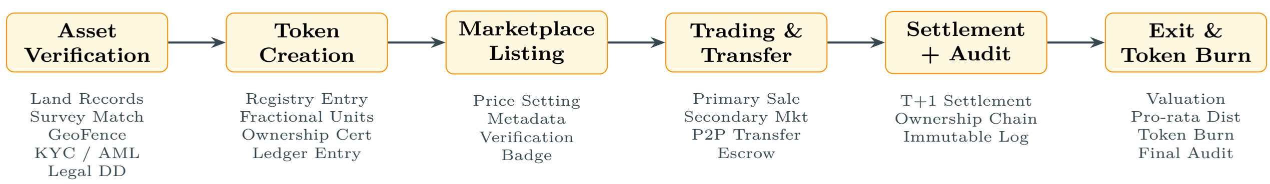

Figure 4: Land Tokenization Lifecycle — Six-Phase Workflow.

9.4 Marketplace Engine

Attribute	Description
Purpose	Production marketplace for agricultural investment listings, token trading, and P2P transactions.
Inputs	Listings (farm, token, collective vehicle), buy/sell orders, search queries, watchlist actions.
Outputs	Matched orders, settlement instructions, portfolio updates, fee calculations, tax withholding records.
Internal Logic	**Listing:** Verified assets listed with metadata, pricing (fixed / auction / instant buy), and availability. **Search & Discovery:** Full-text search + faceted filters (crop, state, risk level, return range, ESG score, liquidity). Ranking by relevance $\times$ opportunity score. **Order Matching:** For auctions: sealed-bid second-price (Vickrey) or English ascending. For fixed price: FIFO matching. For instant buy: immediate execution at listed price. **Escrow:** All trades pass through escrow service. Buyer funds $\rightarrow$ escrow $\rightarrow$ ownership transfer $\rightarrow$ funds release. Timeout: 48h auto-cancel if settlement fails. **Fees:** Platform fee (configurable, default 1.5% of transaction value). Tax withholding (TDS) computed per regulatory rules. **Post-Trade:** Buyer's portfolio updated. Seller's holdings decremented. Notification to both parties.
Dependencies	Tokenization Engine, Escrow Service, Verification Pipeline, Notification Service.
Storage	PostgreSQL (`listings`, `orders`, `trades`, `watchlists`).
Failure Modes	Order matching engine overload (queue with back-pressure), escrow timeout (auto-refund).
Update Cadence	Real-time (event-driven).

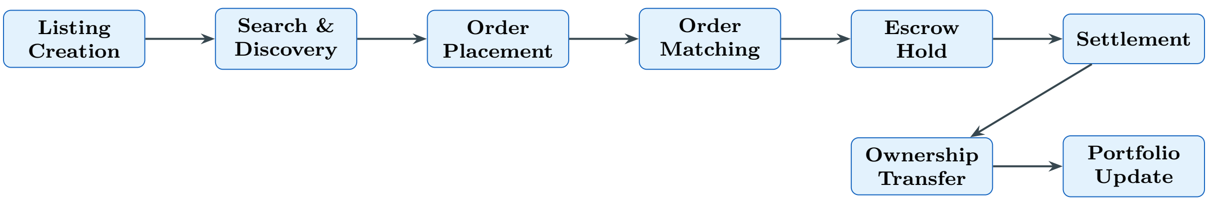

Figure 5: Marketplace Engine — Order Lifecycle.

10 Verification Pipeline

Every stakeholder undergoes multi-stage verification. The pipeline is modular: each verification step is an independent microservice that returns a pass/fail/pending result with confidence score.

10.1 Farmer Verification

#	Step	Method	Outcome
1	Phone OTP	SMS gateway	Identity binding
2	Aadhaar Verification	UIDAI API (hash only)	Legal identity
3	PAN Verification	NSDL/UTI API	Tax identity
4	Face Match	Liveness + Aadhaar photo match	Anti-impersonation
5	Bank Verification	Penny drop test	Account ownership
6	Land Record Validation	State revenue API	Ownership proof
7	GeoFence Validation	PostGIS polygon match	Spatial accuracy
8	Satellite Validation	NDVI $>$ 0.3 trailing 12m	Agricultural use
9	Historical Crop Check	Crop history + satellite	Farming activity
10	Farm Inspection	Inspector app (optional)	Physical validation

10.2 Investor Verification

#	Step	Method	Outcome
1	KYC (PAN + Aadhaar)	NSDL + UIDAI API	Legal identity
2	Bank Verification	Penny drop + IFSC	Account ownership
3	AML Screening	Sanctions list + PEP check	Compliance
4	Risk Profile	Questionnaire + scoring	Risk appetite
5	Income Verification	ITR / bank statement OCR	Eligibility
6	Investment Eligibility	Accreditation rules	Regulatory
7	Face Match	Liveness detection	Anti-fraud
8	Email Verification	OTP	Contact binding

10.3 Expert & Insurance Provider Verification

#	Step	Method	Outcome
1	Identity (KYC)	PAN + Aadhaar	Legal identity
2	Degree / License	DigiLocker API / manual	Qualification
3	Experience	Employment verification	Competence
4	Organisation	MCA / registrar lookup	Entity validation
5	Regulatory	License number validation	Compliance

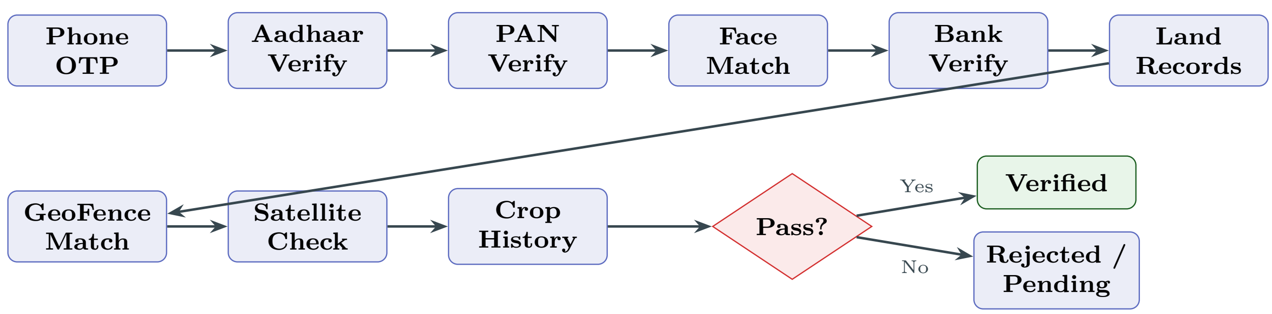

Figure 6: Farmer Verification Pipeline — Sequential Multi-Stage Workflow.

11 Data Flow Diagrams

11.1 Level 0 — Context Diagram

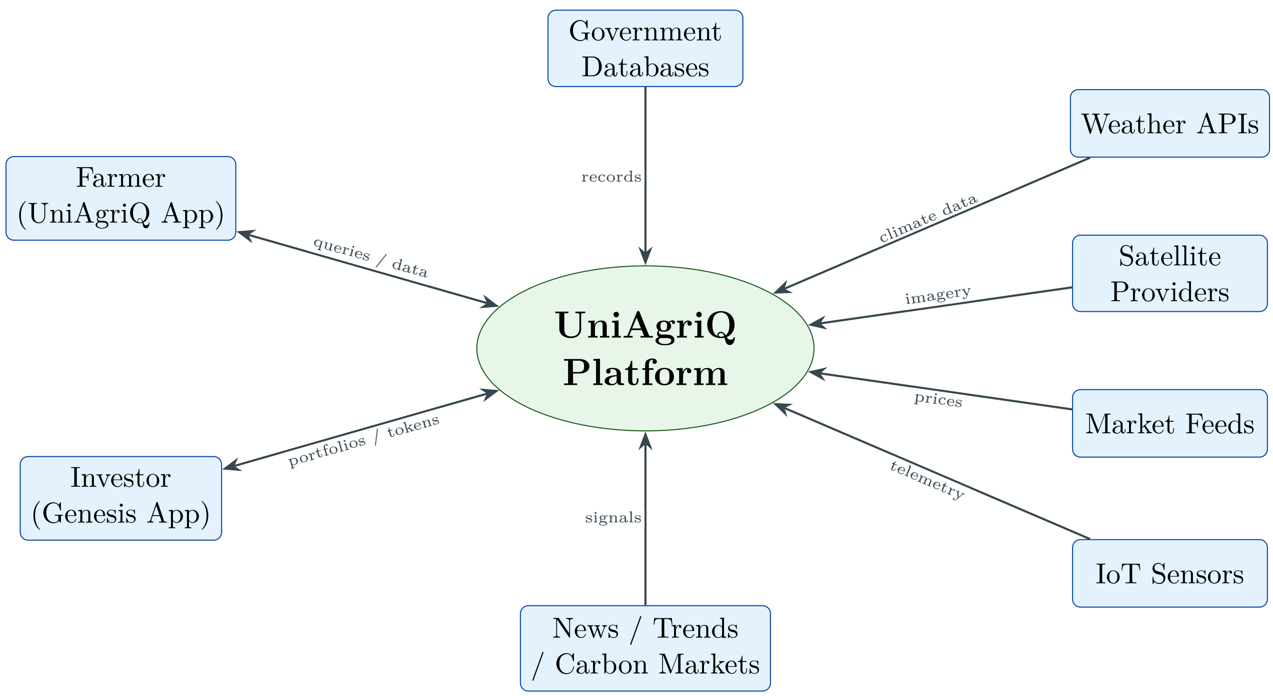

Figure 7: Level 0 Data Flow Diagram — System Context.

11.2 Level 1 — Subsystem Interactions

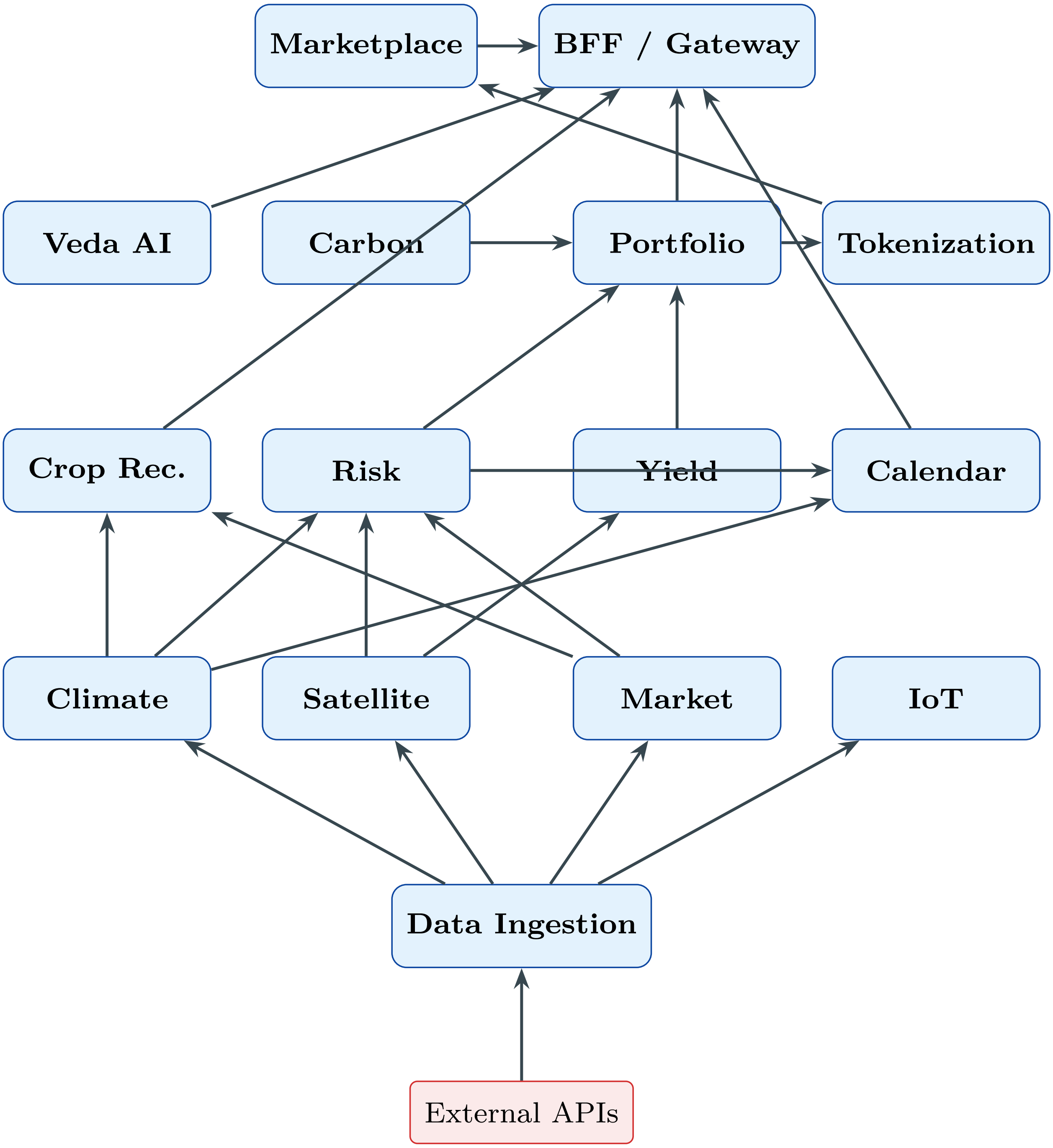

Figure 8: Level 1 Data Flow — Subsystem Interactions.

12 Use Case Diagrams

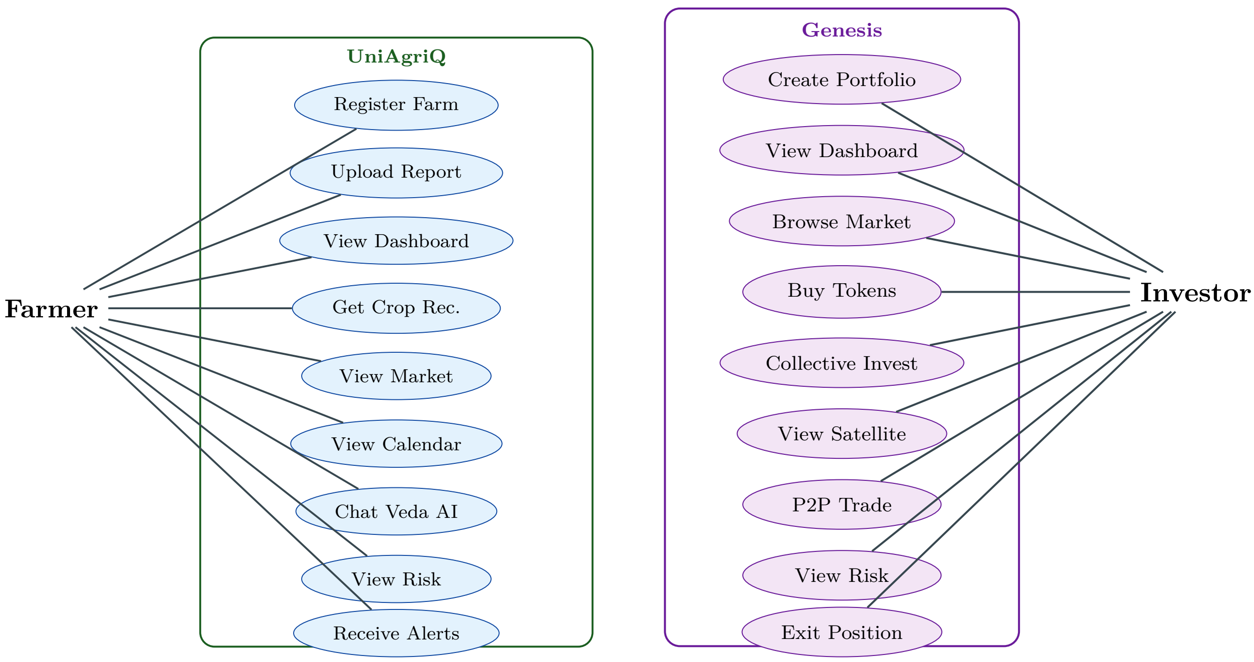

Figure 9: Use Case Diagrams — UniAgriQ (Farmer) and Genesis (Investor).

13 Component Diagram

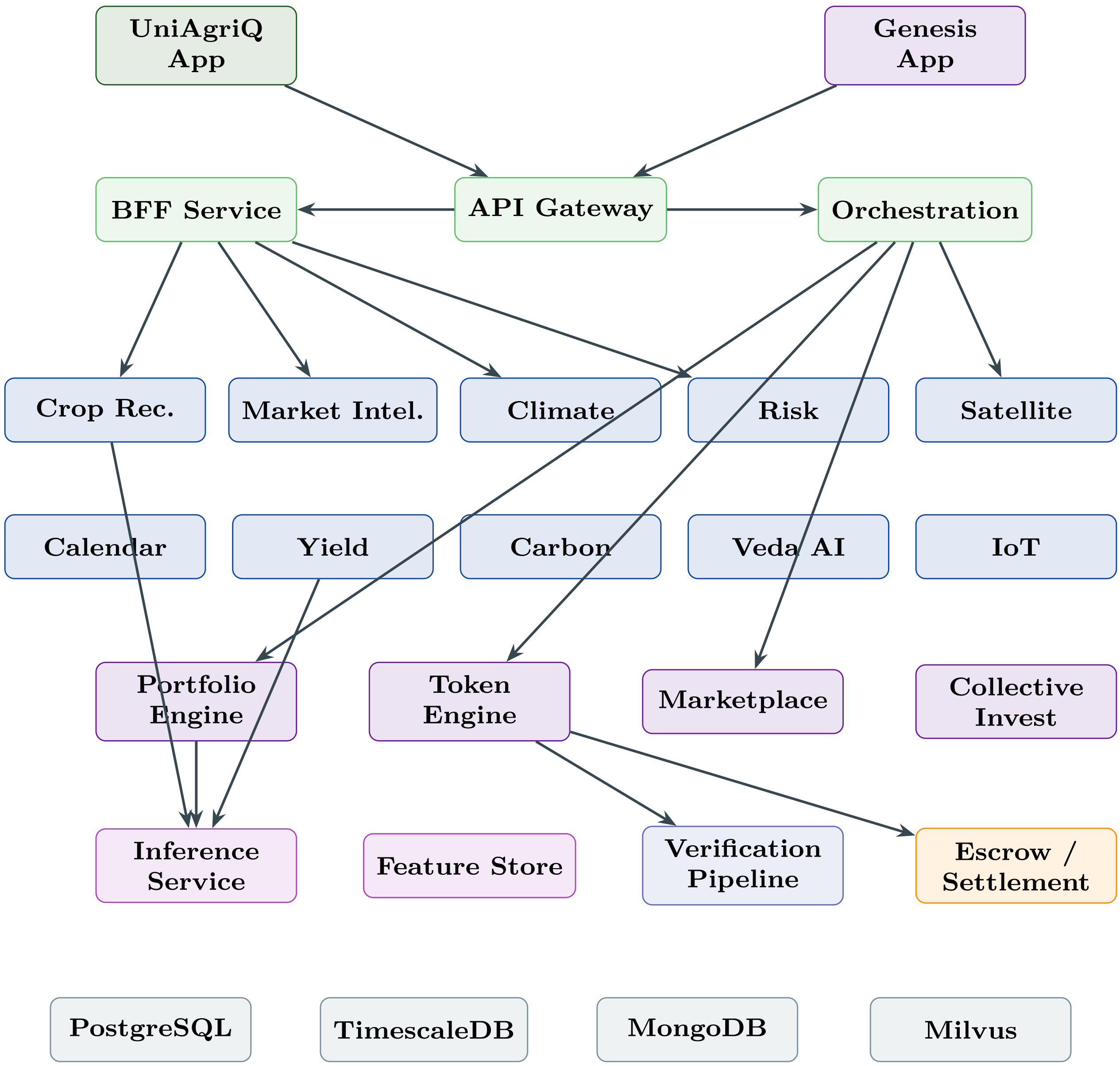

Figure 10: Component Diagram — Full UniAgriQ + Genesis Topology.

14 Deployment Diagram

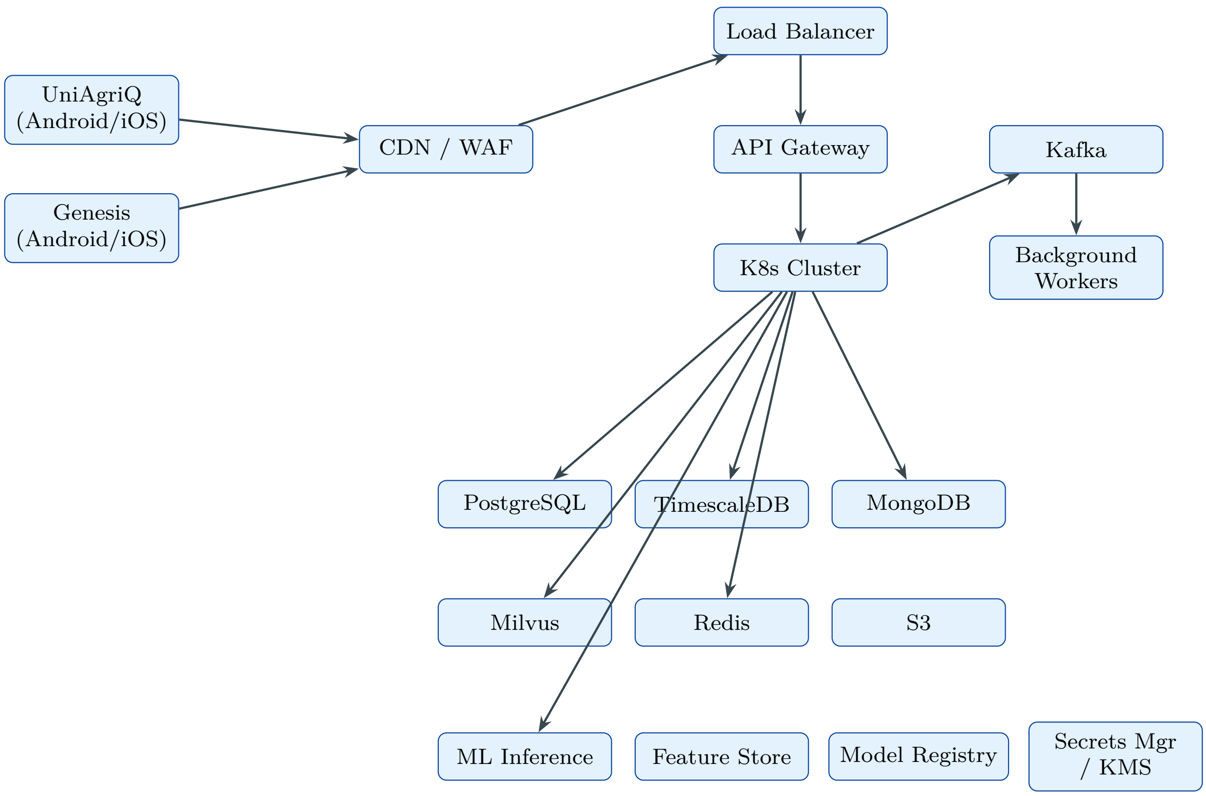

Figure 11: Deployment Diagram — Cloud Infrastructure.

15 Database & Data Store Architecture

UniAgriQ employs **polyglot persistence** with logically separated databases per domain.

15.1 Storage Engine Allocation

Database	**Engine**	**Tables / Collections**
User Database	PostgreSQL	`farmers`, `investors`, `experts`, `verification_status`
Farm Database	PostgreSQL + PostGIS	`farms`, `farm_polygons`, `crop_history`, `farm_ownership`
Portfolio Database	PostgreSQL	`portfolios`, `portfolio_positions`, `collective_vehicles`, `cap_table`, `distributions`
Token Database	PostgreSQL	`token_registry`, `ownership_ledger`, `token_trades`, `token_burns`
Market Database	PostgreSQL + TimescaleDB	`listings`, `orders`, `trades`, `mandi_prices` (hypertable), `commodity_prices` (hypertable)

Climate Database	TimescaleDB	`climate_observations`, `climate_forecasts` (hypertables)
Satellite Database	TimescaleDB + S3	`satellite_indices` (hypertable), imagery tiles in S3
IoT Database	TimescaleDB	`sensor_readings` (hypertable)
Calendar Database	PostgreSQL	`crop_calendars`, `calendar_tasks`
Notification DB	PostgreSQL	`notifications`, `notification_preferences`
Knowledge Base	MongoDB	`crop_knowledge`, `disease_knowledge`, `soil_reports`, `news_articles`
Vector Database	Milvus	`knowledge_embeddings`, `chat_embeddings`
Audit Logs	PostgreSQL	`audit_log`, `transaction_ledger` (immutable)
Model Outputs	PostgreSQL	`recommendations`, `risk_scores`, `yield_predictions`, `carbon_estimates`, `investor_scores`
Document Storage	S3	Reports, certificates, legal documents
Cold Storage	S3 Glacier	Archived imagery, old logs
Cache	Redis Cluster	Session tokens, latest readings, BFF cache

15.2 Entity Relationship Overview

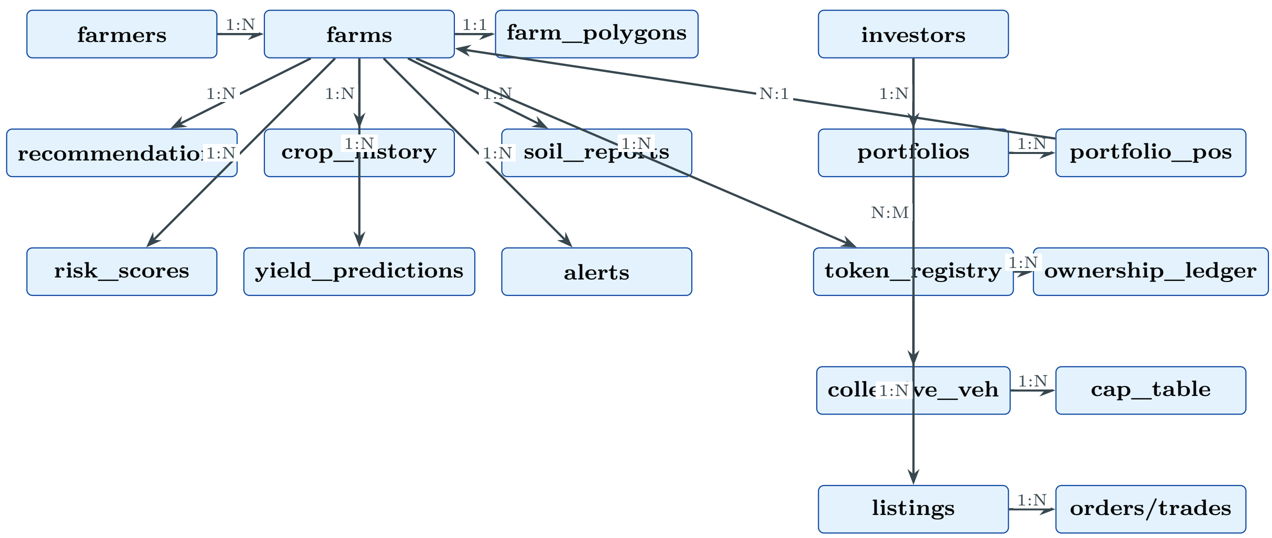

Figure 12: Entity Relationship Diagram — Farm + Investment Domains.

16 API Integration Architecture

16.1 External API Inventory

Category	Provider	Protocol	Cadence	Fallback
Weather	OpenWeatherMap	REST	15 min	Tomorrow.io
Forecast	IMD API	REST	6 hr	OpenWeatherMap
Satellite	Sentinel Hub	REST	5 days	Earth Engine

Mandi	Agmarknet	REST	1 hr	data.gov.in
Commodity	MCX/NCDEX	WebSocket	Real-time	Delayed REST
Import-Export	UN Comtrade	REST	Monthly	Manual upload
News	NewsAPI/GDELT	REST	1 hr	RSS parser
KYC	UIDAI + NSDL	REST	On-demand	Manual review
AML	Sanctions API	REST	On-demand	Manual check
Bank Verify	Penny drop API	REST	On-demand	Manual
OCR	Google Vision	REST	Per-upload	Tesseract
LLM	OpenAI/Google	REST	Per-query	Self-hosted
IoT	AWS IoT Core	MQTT	Real-time	Mosquitto
Carbon Price	Market API	REST	Daily	Manual config

16.2 Key Internal API Endpoints

Endpoint	Verb	Description
`/farms/{id}/health`	GET	Farm health dashboard
`/farms/{id}/recommendations`	POST	Trigger crop recommendation
`/farms/{id}/calendar`	GET	Crop calendar
`/farms/{id}/risk`	GET	Risk scores
`/farms/{id}/yield`	GET	Yield prediction
`/farms/{id}/satellite`	GET	Satellite indices
`/market/{crop}/intelligence`	GET	Market intelligence
`/veda/chat`	POST	Veda AI turn
`/portfolio/create`	POST	AI portfolio construction
`/portfolio/{id}/rebalance`	POST	Rebalancing trigger
`/tokens/{asset}/create`	POST	Tokenize asset
`/tokens/{id}/transfer`	POST	Token transfer
`/marketplace/listings`	GET	Browse marketplace
`/marketplace/orders`	POST	Place order
`/collective/{id}/subscribe`	POST	Join collective vehicle
`/verification/{id}/status`	GET	Verification status
`/wallet/{id}/balance`	GET	Wallet balance
`/escrow/{id}/status`	GET	Escrow status

17 AI / ML Pipeline Architecture

17.1 Pipeline Overview

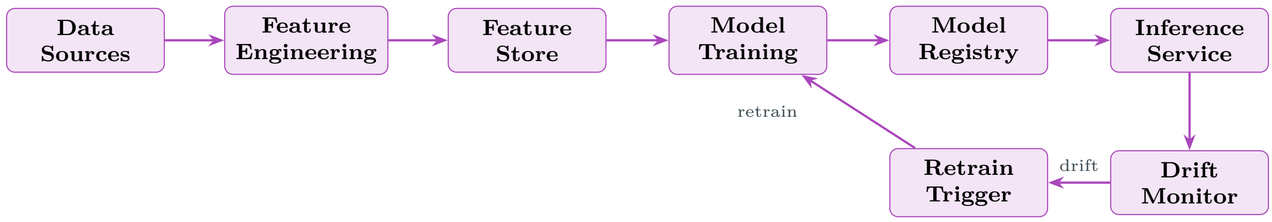

Figure 13: ML Pipeline — Training and Inference Lifecycle.

17.2 Model Inventory

Model	Type	Inputs	Retrain	Metric
Crop Suitability	XGBoost	Soil, climate	Quarterly	NDCG@10
Yield Regression	LightGBM	Soil, NDVI, GDD	Seasonal	RMSE
Demand Forecaster	LSTM/Prophet	Price, season	Monthly	MAPE
News Sentiment	BERT (fine-tuned)	Article text	Quarterly	F1
Disease Risk	Random Forest	Weather, stage	Quarterly	AUC-ROC
Soil Report NER	Custom NER	OCR text	Biannual	Entity F1
Portfolio Optimizer	CVXPy/Scipy	Returns, risk	Monthly	Sharpe
Fraud Detector	Autoencoder	Transaction features	Monthly	Precision
Price Forecaster	Transformer	Price series	Monthly	MAPE
Climate Forecaster	ConvLSTM	Gridded weather	Seasonal	MAE
Satellite Analyser	U-Net	Imagery bands	Biannual	IoU
Recommendation	Two-tower	User + farm features	Weekly	CTR
Embedding Model	Sentence-BERT	Text chunks	On corpus	Recall@5
Intent Classifier	DistilBERT	Chat queries	Monthly	Accuracy

17.3 Drift Monitoring

- **Data drift:** Kolmogorov–Smirnov test on feature distributions (weekly). Alert if $p < 0.01$ on >3 features.
- **Concept drift:** Rolling evaluation against ground truth. Alert if metric degrades >10%.
- **Prediction drift:** Population stability index (PSI) on outputs. Alert if PSI > 0.2.

18 Scoring Formula Architecture

All scores follow:

$$S_{\text{composite}} = \sum_{i=1}^{n} w_i \cdot \phi_i(x_i), \quad \sum w_i = 1, \quad \phi_i : \mathbb{R} \to [0,1] \tag{1}$$

18.1 Farm Health Score

$$S_{\text{health}} = 0.25\,\phi(\Delta\text{NDVI}) + 0.20\,\phi(\text{soil}) + 0.15\,\phi(\text{sensor}) + 0.25\,(1 - \phi(\text{risk})) + 0.15\,\phi(\text{calendar\_adherence}) \tag{2}$$

Colour: $S < 0.40 \Rightarrow$ Red, $0.40 \leq S < 0.70 \Rightarrow$ Blue, $S \geq 0.70 \Rightarrow$ Green.

18.2 Crop Suitability Score

$$S_{\text{crop}} = 0.25\,S_s + 0.20\,S_c + 0.15\,S_r + 0.25\,S_m - 0.15\,S_d \tag{3}$$

Rejection: $S_{\text{crop}} < 0.45$. Dimension floor: any < 0.20 triggers high-risk flag.

18.3 Soil Suitability (S_s)

$$S_s = 1 - \frac{1}{k}\sum_{j=1}^{k} \min\left(1, \frac{|x_j - \mu_j^{\text{opt}}|}{\sigma_j^{\text{tol}}}\right) \tag{4}$$

18.4 Climate Fit (S_c)

$$S_c = \prod_{p \in \{\text{temp, humid, solar}\}} \exp\left(-\frac{(x_p - \mu_p^{\text{opt}})^2}{2(\sigma_p^{\text{tol}})^2}\right) \tag{5}$$

18.5 Market Profit Score (S_m)

$$S_m = \phi(\text{price\_trend}) \times \phi(\text{demand}) \times (1 - \phi(\text{volatility})) \tag{6}$$

18.6 Yield Confidence Score

$$S_{\text{yield}} = 1 - \frac{Q_{90} - Q_{10}}{2 \cdot Q_{50}} \tag{7}$$

18.7 Investor Score

$$S_{\text{investor}} = [0.20,\ 0.20,\ 0.15,\ 0.15,\ 0.10,\ 0.10,\ 0.10] \cdot [S_{\text{health}},\ 1 - S_{\text{risk}},\ S_{\text{yield}},\ S_{\text{sat}},\ S_{\text{clim}},\ S_{\text{carbon}},\ S_{\text{opp}}]^T \tag{8}$$

18.8 Portfolio Diversification Score

$$S_{\text{div}} = 1 - \frac{\bar{\rho}(\mathbf{w})}{\rho_{\max}} \tag{9}$$

where $\bar{\rho}(\mathbf{w})$ is the weighted average pairwise correlation across portfolio positions and $\rho_{\max}$ is the theoretical maximum.

18.9 ESG / Sustainability Score

$$S_{\text{ESG}} = 0.40\, S_{\text{carbon}} + 0.25\, \phi(\text{water\_eff}) + 0.20\, \phi(\text{biodiversity}) + 0.15\, \phi(\text{social\_impact}) \tag{10}$$

19 Mobile App Information Architecture

19.1 UniAgriQ — Farmer App

Bottom navigation: **Home**, **Farm**, **Market**, **Calendar**, **Profile**. Floating: **Veda AI**.

Home Screen (vertical scroll):

1. Alert banner (horizontally scrollable chips; red = emergency).
2. Farm Health Card (glassmorphism; colour-coded score circle, crop name, growth stage badge, mini geofence preview).
3. Multi-crop summary row (horizontal chip scroll).
4. Upcoming activities (next 3 calendar tasks, colour-coded).
5. Quick actions grid (2×3): Get Recommendations, Upload Report, View Calendar, Market Prices, Ask Veda AI, Risk Check.
6. Advisory feed (urgency-ranked cards).
7. Opportunity cards (horizontal scroll).
8. News slider (auto-scrolling headlines).
9. Sensor summary bar (if IoT connected).

19.2 Genesis — Investor App

Bottom navigation: **Home**, **Portfolio**, **Market**, **Discover**, **Profile**.

Genesis Home Screen is designed to feel like a premium fintech application (comparable to Zerodha, Groww, INDmoney, Robinhood) where the underlying assets are agricultural investments.

Home Screen cards (vertical scroll):

1. **Portfolio Value Card:** Total portfolio value, today's gain/loss (absolute + %), expected annual return, portfolio performance sparkline.

2. **Score Cards Row:** Risk Score (gauge), Diversification Score (gauge), Farm Health Index (colour-coded), ESG Score.
3. **Investment Wallet:** Balance, pending deposits, recent transactions (last 3).
4. **Portfolio Performance Graph:** Interactive line chart (1W/1M/3M/1Y/ALL toggles), benchmark comparison.
5. **Investment Allocation Pie Chart:** By crop type, by state, by risk level (toggle).
6. **AI Suggestions Card:** “Rebalance recommended”, “New high-opportunity farm available”, “Harvest revenue incoming”.
7. **Active Investments:** List of current positions with farm thumbnail, crop, return %, health dot.
8. **New Investment Opportunities:** Horizontally scrollable opportunity cards with satellite preview, yield forecast, risk badge.
9. **Harvest Timeline:** Upcoming harvest dates across portfolio with expected revenue.
10. **Profit Calendar:** Month-wise expected profit distribution heatmap.
11. **Carbon Credit Value:** Total carbon credits earned, current market value, trend.
12. **Weather Exposure:** Map showing weather risk overlay across portfolio farms.
13. **Trending Crops & High Demand Commodities:** Market momentum indicators.
14. **Opportunity Alerts:** Supply gap, price spike, new collective vehicle opening.
15. **Upcoming Exit Opportunities:** Positions approaching maturity or exit window.
16. **P2P Marketplace Activity:** Recent token trades, volume, price changes.
17. **Insurance Coverage:** Portfolio-level insurance status.
18. **Pending Verification:** Assets awaiting verification completion.
19. **Global Market Exposure:** International commodity price impact on portfolio.
20. **Liquidity Index:** Portfolio-level liquidity assessment based on secondary market depth.
21. **Urgent Alerts:** Risk threshold breaches, weather emergencies, regulatory changes.
22. **Recommended Portfolio Templates:** Conservative, Balanced, Growth, ESG, High-Yield pre-built options.
23. **Recent Transactions:** Full transaction history with status badges.

Genesis UI Principles:

- Dark mode primary with green/purple accent palette.
- Glassmorphism cards with frosted-glass effect.

- Financial data presented with sparklines, gauges, and colour-coded deltas (green = gain, red = loss).
- Inter or SF Pro font family.
- Haptic feedback on trade actions.
- Bottom sheet for quick trade/buy flows.
- Pull-to-refresh with portfolio value animation.

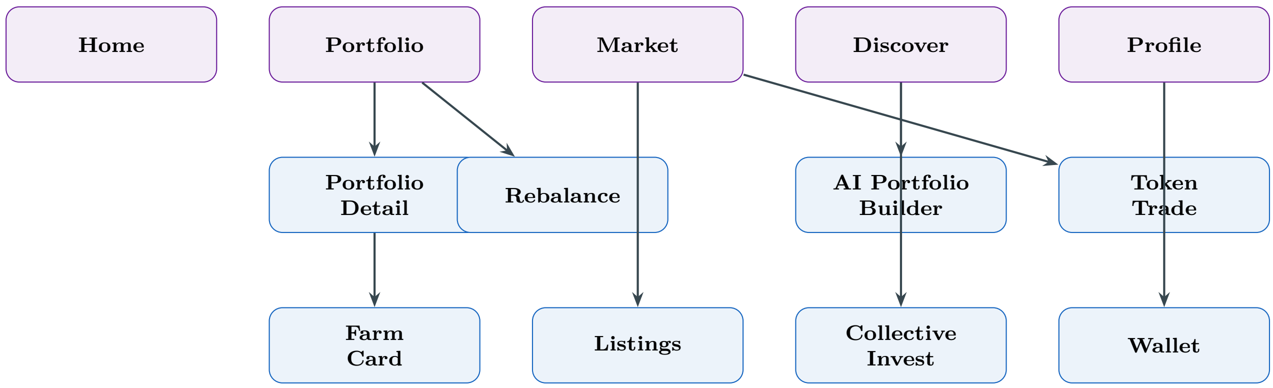

Figure 14: Genesis App — Navigation Architecture.

20 Security & Access Control Model

20.1 Authentication

- **UniAgriQ:** Phone + OTP → JWT (15 min access + 30 day refresh). Optional biometric.
- **Genesis:** Email + password + TOTP MFA → JWT. SSO via Google/Microsoft. Biometric for trade confirmation.
- **Services:** mTLS between microservices.
- **External APIs:** Keys in AWS Secrets Manager, rotated quarterly.

20.2 Authorisation

RBAC + ABAC hybrid. Roles define baseline permissions; attribute-based policies add fine-grained control (e.g., investor can only view farms in their portfolio).

Role	Permissions
Farmer	CRUD own farm; read own scores/calendar/alerts; chat Veda AI.
Investor	Read portfolio farms; trade tokens; create portfolios; P2P.
Admin	Full CRUD; configure weights; manage users; audit logs.
System	Execute batch jobs; write all stores; invoke inference.
Veda AI	Read farm profile, knowledge base, engine outputs.

20.3 Data Protection

- AES-256 at rest. TLS 1.3 in transit. Column-level encryption for PII.
- KMS (AWS KMS) for key management with automatic rotation.
- Aadhaar: SHA-256 hash only. Phone: envelope encryption.
- No PII in logs (structured logging with scrubbing).
- India-region cloud (ap-south-1).

20.4 Financial Security

- **Escrow protection:** Funds held in regulated escrow account; released only on settlement confirmation.
- **Transaction monitoring:** Real-time velocity checks, amount anomaly detection, geographic anomaly detection.
- **Wallet security:** PIN + biometric for withdrawals. Daily withdrawal limits configurable per risk profile.
- **Fraud detection AI:** Autoencoder-based anomaly detector on transaction features (amount, frequency, counterparty, time-of-day). Alert on reconstruction error $> 3\sigma$.

20.5 Zero Trust Architecture

- No implicit trust between services; every request authenticated.
- Network segmentation: compute, data, and ML tiers in separate VPC subnets.
- API rate limiting: per-user, per-service, per-endpoint quotas.
- Audit log: every data mutation recorded with actor, timestamp, resource, old/new value hash.

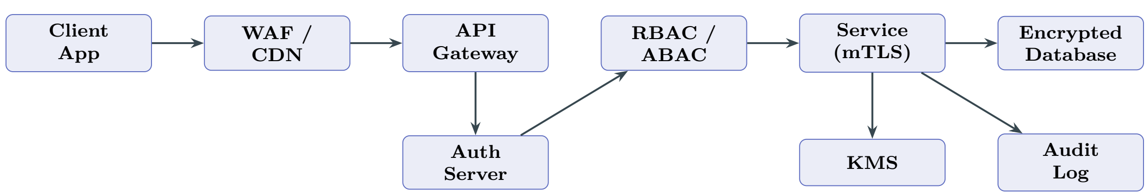

Figure 15: Security Architecture — Zero Trust Request Flow.

21 Observability & Monitoring

21.1 Three Pillars

Pillar	Tool	Scope	Retention
Logging	ELK / Loki	Structured JSON from all services	90d hot, 1y cold
Metrics	Prometheus + Grafana	Latency, throughput, errors, queue depth, inference time	15 months
Tracing	Jaeger / Tempo	Distributed request traces	30 days

21.2 Key Alerts

Metric	Threshold	Action
API p95 latency	> 500 ms for 5 min	PagerDuty P2
5xx error rate	$> 1\%$ for 5 min	PagerDuty P1
Pipeline delay	$> 2\times$ cadence	Slack + P3
ML inference	> 3 s p95	Slack + P3
Model drift (PSI)	> 0.2	ML team + Jira
External API fail	>3 consecutive	Circuit breaker + Slack
Escrow timeout	>48h unsettled	Finance team + P2
Fraud score	> 0.85	Security team + block

22 Scalability & Fault Tolerance

22.1 Horizontal Scaling

Component	Mechanism
API Gateway	K8s HPA (CPU >60%); ALB with multiple targets.
Intelligence Engines	Stateless pods; independent HPA per engine.
Portfolio Engine	Compute-intensive; dedicated node pool with autoscaler.
Marketplace	Order matching: single-writer with event-sourcing for consistency; read replicas for browsing.
ML Inference	SageMaker auto-scaling; GPU for embedding/LLM.
Kafka	Partition scaling for high-throughput topics.
PostgreSQL	Read replicas (up to 5); PgBouncer connection pool.
TimescaleDB	Compression + continuous aggregates; multi-node >1TB.
Redis	Cluster mode (3 primaries + replicas).

22.2 Fault Tolerance

- **Circuit breaker:** All external calls wrapped; open after 5 failures in 60s; half-open after 30s.
- **Retry:** Exponential backoff (200ms base, 3 max, jitter).
- **Dead-letter queue:** Failed ingestion to DLQ; auto-retry after 1h.
- **Graceful degradation:** Non-critical engine down → partial dashboard with placeholder.
- **Multi-AZ:** All stateful components across ≥2 AZs.
- **DB failover:** Streaming replication with automatic failover (RDS Multi-AZ or Patroni).

23 Backend Infrastructure Services

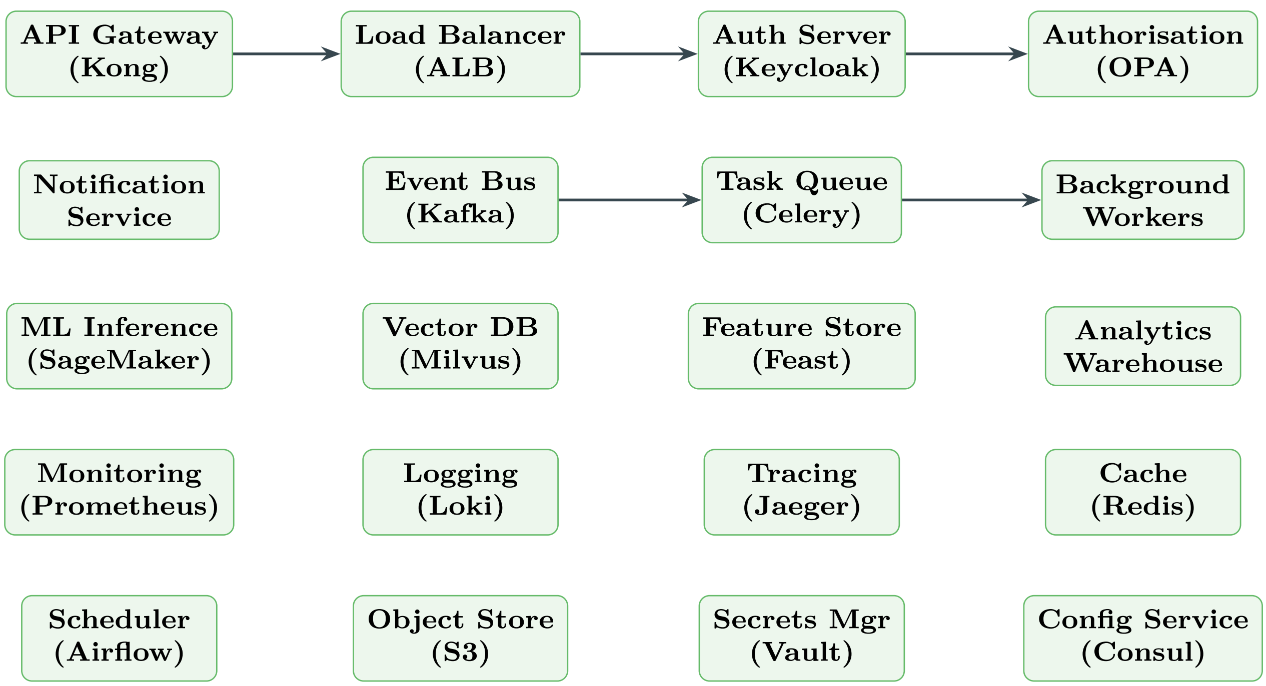

Figure 16: Backend Infrastructure — Service Inventory.

Appendix: Detailed Engine Diagrams

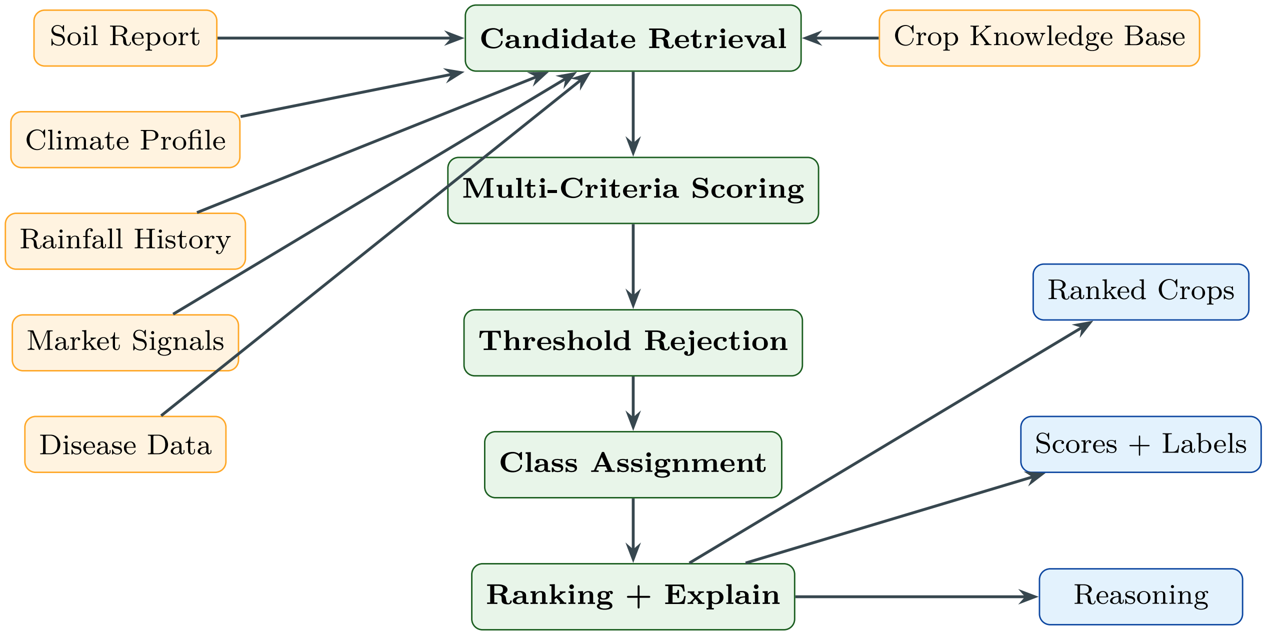

Figure 17: Crop Recommendation Engine Pipeline.

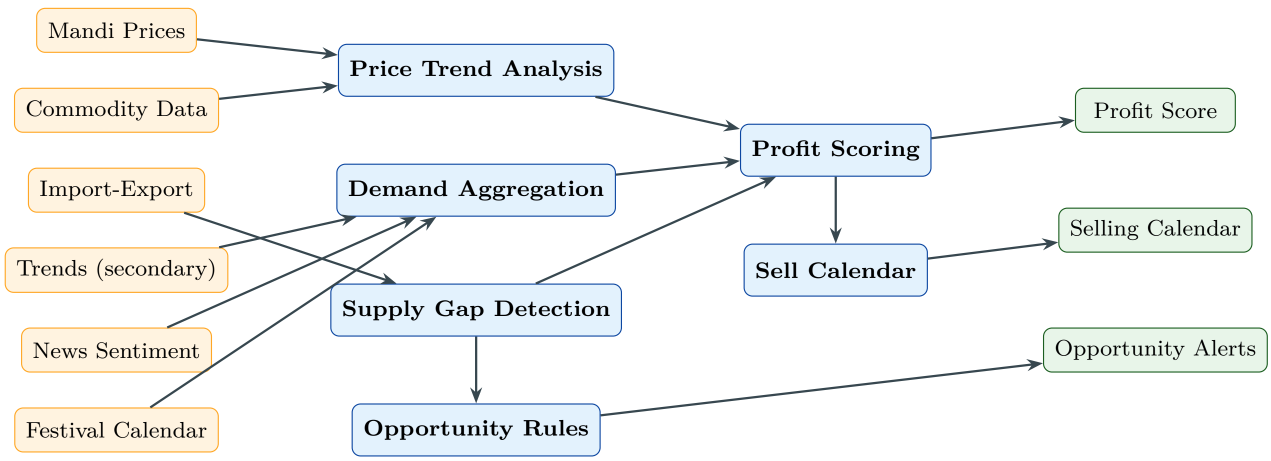

Figure 18: Market Intelligence Engine Pipeline.

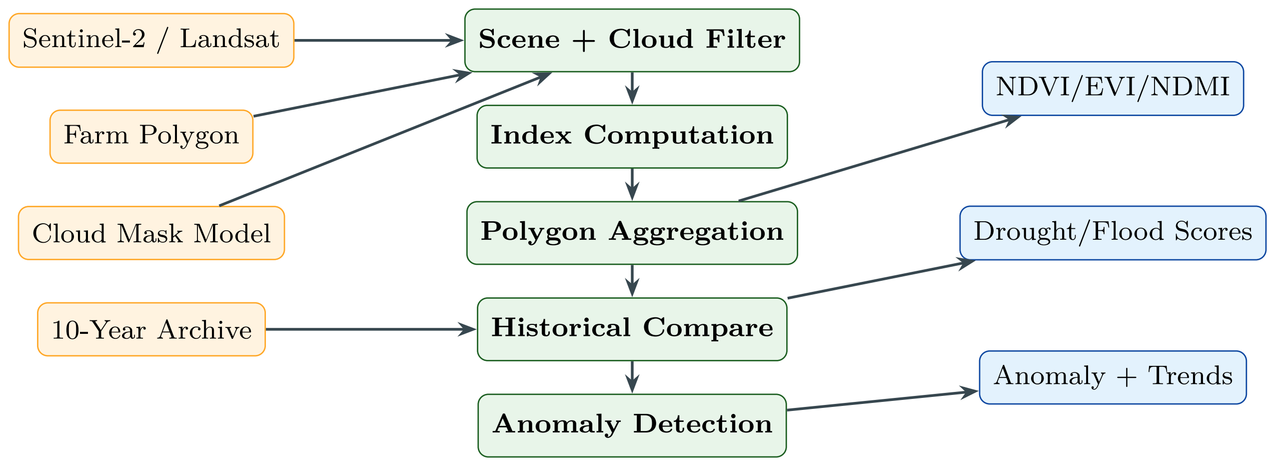

Figure 19: Satellite Intelligence Engine Pipeline.

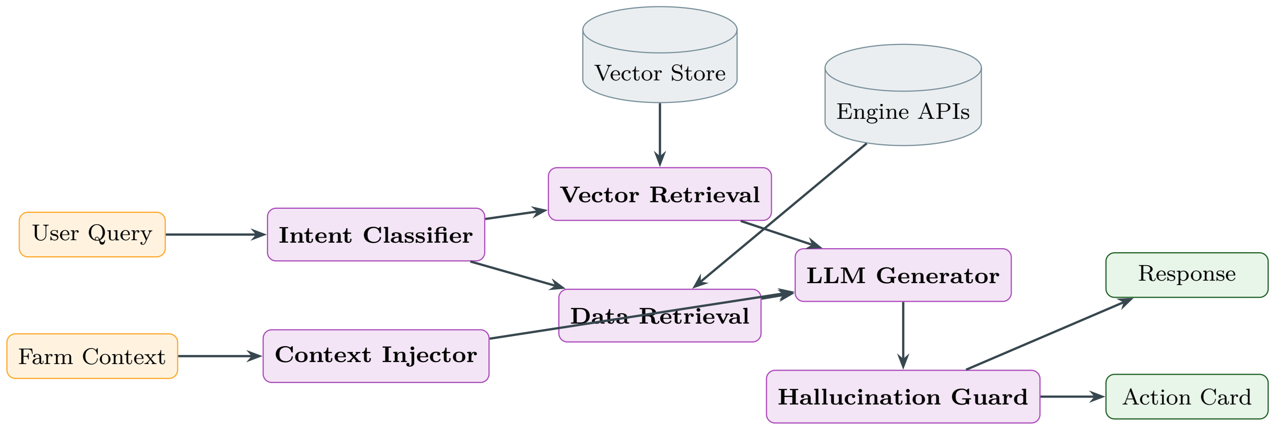

Figure 20: Veda AI — RAG Architecture.

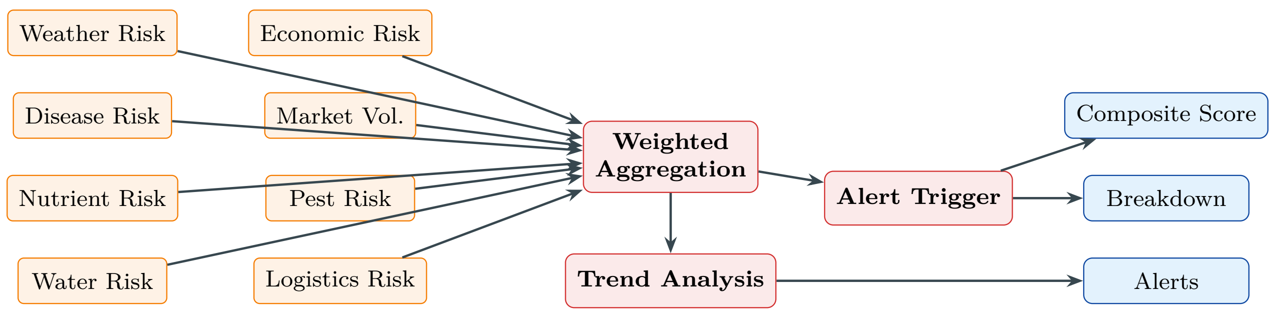

Figure 21: Risk Engine — 8-Dimension Assessment.

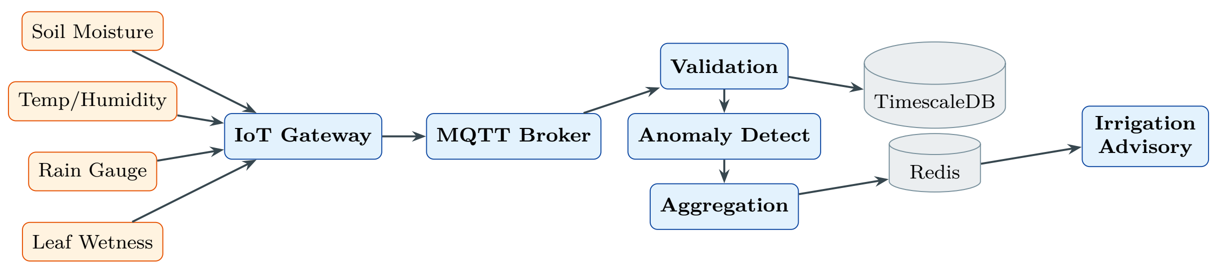

Figure 22: IoT Sensor Data Flow Pipeline.

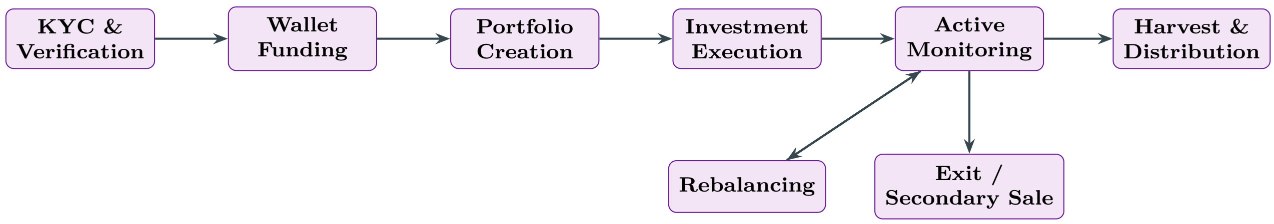

Figure 23: Genesis — Investor Lifecycle.

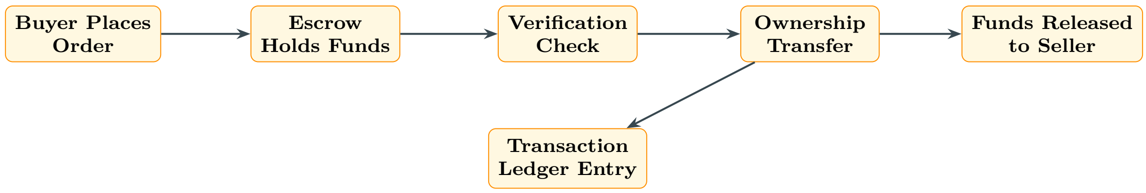

Figure 24: Escrow-Based Settlement Flow.

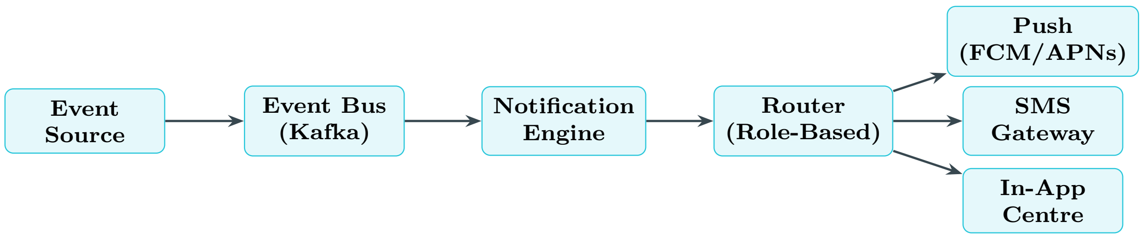

Figure 25: Notification Pipeline — Event-Driven Routing.

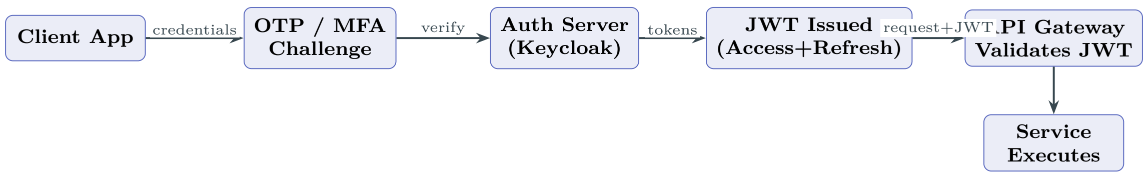

Figure 26: Authentication Flow — OTP/MFA to JWT.

24 Conclusion

This document specifies the complete system architecture of the UniAgriQ platform comprising the **UniAgriQ** farmer application and the **Genesis** investor application, unified by a shared intelligent backend. The architecture is characterised by:

1. **Dual-application topology:** Two independent mobile applications serving fundamentally different user personas while sharing AI engines, market intelligence, satellite analytics, risk computation, authentication, notification infrastructure, and polyglot databases.

2. **AI Portfolio Engine:** A novel agricultural portfolio construction system that treats farm investments like mutual fund units — diversified across crop type, geography, climate zone, harvest season, water dependency, and disease exposure using constrained portfolio optimisation.

3. **Land Tokenization:** A six-phase tokenization lifecycle from multi-stage asset verification through fractional unit creation, marketplace listing, escrow-based settlement, and token burn on exit — with immutable ownership ledger and fraud detection.

4. **Collective Investment:** Multi-investor vehicles with automated CAP table management, pro-rata profit distribution, weighted governance, and multiple exit mechanisms.

5. **Production Marketplace:** Full order lifecycle with listings, search, auction/fixed-price/instant-buy modes, order matching, escrow settlement, and portfolio reconciliation.

6. **Multi-stage Verification:** Ten-step farmer verification and eight-step investor verification combining government API validation, biometric matching, satellite confirmation, and optional physical inspection.

7. **Zero hard-coded data:** Every score, recommendation, alert, calendar entry, portfolio allocation, and token valuation derived from live ingestion, model inference, rule-engine evaluation, or scheduled computation.

8. **Eighteen subsystems** with explicit input-output contracts, dedicated storage, and defined failure modes — spanning agricultural intelligence, financial services, and marketplace operations.

9. **Comprehensive security:** Zero-trust architecture with OAuth 2.0, RBAC+ABAC, KMS, fraud detection AI, escrow protection, transaction monitoring, and immutable audit trails.

10. **Production-grade operations:** Twelve-layer architecture, horizontal scaling strategy, circuit breakers, multi-AZ deployment, polyglot persistence across seven storage engines, and a complete observability stack.

UniAgriQ and Genesis together represent a convergence of precision agriculture, agricultural finance, market intelligence, climate science, and AI-driven decision support into a unified platform that transforms both farming operations and agricultural investment.

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